

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 35

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Organic (methyl) mercury	1
01.2	• Anaerobic microbes converted the inorganic mercury into organic methyl mercury • Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people	2
01.3	• Most heavy metals are less soluble at high pH • So they remain immobile and are not leached into water or absorbed by organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
02.1	• Detritivores (worms, beetle larvae, millipedes, woodlice) physically break up dead organic matter, increasing its surface area, and release nutrients; worms create tunnels improving drainage and aeration • Decomposers (bacteria, fungi) secrete enzymes that break down dead organic matter, releasing nutrient ions (ammonification) and forming humus which binds particles into crumb peds • Nitrogen-fixing bacteria convert nitrogen gas to ammonium ions; nitrifying bacteria oxidise ammonium to nitrite then nitrate for root absorption • Mycorrhizal fungi form symbiotic associations with roots increasing phosphate uptake	6
02.2	• This is the range of tolerance of most plants and soil biota; acidic soils leach nutrients and mobilise toxic aluminium, damaging root membranes • In alkaline soils phosphates become insoluble / unavailable	2
02.3	• Interstitial spaces are reduced so aeration falls (aerobic processes such as nitrification and decomposition slow) and roots cannot penetrate • Infiltration falls so waterlogging is more likely and runoff / erosion increase	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-161; AQA spec 3.2.5.1.	
03.1	<i>Any 4 from:</i> • High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up • High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation) • Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification) • Birds of prey are at the top of the food chain so receive the highest concentrations • Sub-lethal doses caused thin eggshells that broke / reduced breeding success; higher doses caused sterility or death	4
03.2	• Advantage: low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly • Disadvantage: high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase); chronic effects (memory loss, depression); suspected carcinogens	2
03.3	• Pyrethroids are highly toxic to fish (even though mammal toxicity is low)	1
03.4	<i>Any 3 from:</i> • Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic) • They are moderately persistent so remain in the soil long enough to be absorbed • Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats • So bees / pollinators are less exposed than when the crop is sprayed	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.	
04.1	• Acidophilic bacteria (eg Acidithiobacillus) oxidise sulfide ores, producing sulfuric acid which dissolves the metal ions • Leachate (rainwater / recirculated solution) percolating through the spoil heap is collected; recirculation increases the metal concentration • Copper is recovered from solution by electrolysis or by displacement with scrap iron (iron is more reactive so copper is deposited as solid metal) • Smelting needs chemical reduction at high temperatures; energy needed rises rapidly as ore grade falls, so low-grade ores are uneconomic • Bioleaching works at low temperatures with low energy inputs, so it can exploit low-grade deposits and wastes (increasing reserves) - mimics natural low-temperature processes	5
04.2	• Plants (hyperaccumulators) absorb metal ions from soil / water and concentrate them in their leaves • The vegetation is harvested and incinerated • The concentrated metal in the ash is dissolved in acid and separated by electrolysis; also decontaminates polluted sites	3
04.3	• Uranium ions dissolved in seawater adsorb onto the surface of synthetic or natural polymers (lignin from wood, chitin from shrimp shells) • The polymers are collected and the uranium recovered, avoiding mining / high-energy extraction	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 143-145; AQA spec 3.2.3.7.	
05.1	• 7×10^{11} tonnes	1
05.2	• 10% of 10 000 = 1000 billion tonnes • $1000 / 700 \times 100 = 143\%$ (accept 142.9)	2
05.3	• Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) • Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion	2
05.4	• CO ₂ dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis • Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons • When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods	3
05.5	• Aerobic respiration uses oxygen to break down organic compounds completely to CO ₂ (and water), releasing much more energy • Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH ₄), which is later oxidised to CO ₂ in the atmosphere	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	
06.1	• Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them • Built-in obsolescence: products designed to have short lives / wear out / go out of fashion so customers upgrade • Disposable products (razors, lighters, pens) that must be replaced • Over-packaging to make products attractive; also increases wastes from manufacture and mineral extraction	4
06.2	<i>Any 3 from:</i> • Properties of the waste: toxicity, flammability, degradability, radioactivity • Population density (collection costs) • Land availability for landfill • Availability of recycling technology / markets • Legislation / regulatory framework (landfill tax) • Public willingness to recycle / environmental awareness • Processing costs / household income	3
06.3	• Advantage: aerobic decomposition produces little methane; nutrients / organic matter returned to soil (circular economy); reduces landfill volume • Disadvantage: needs separation / collection of clean organic waste; odour / pests / pathogens if poorly managed; low-value product; not suitable for mixed waste	2
06.4	• Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	
07.1	• $(2.9 - 0.7) / 2.9 \times 100 = 75.9\%$ (accept 76%)	1

Question	Marking guidance	Marks
07.2	- Although the larger vehicle uses more energy in total, it carries many more people / a larger load - So the energy per passenger (or per tonne of goods) is lower; steel wheels on rails also have low rolling resistance	2
07.3	<i>Any 3 from:</i> - Flexibility of routes / destinations and door-to-door convenience - Time / speed and reliability where public transport is infrequent - Low fuel tax / cheap fuel - Carrying goods; comfort; status	3
07.4	- Integrated systems (park and ride, bicycles on trains, linked timetables) make it convenient to use the most efficient mode for each part of a journey - So more journeys shift from cars to bulk transport / cycling - ATM / smart motorways use variable speed limits to prevent congestion and keep traffic flowing freely - Vehicles avoid stop-start driving and run nearer their optimum speed (about 56 mph), reducing fuel wasted in acceleration and idling	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-228; AQA spec 3.3.4.2.1.	
08.1	- Readings drift so they may not be accurate; calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions - pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly - If a meter cannot be adjusted, record the percentage error and correct the results	3
08.2	- Same height above the ground - No obstacles to air flow; impeller type facing into the wind - Axis of rotation horizontal (impeller) or vertical (cup); same time period / averaging	3
08.3	- Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling - Evaporation cools the wet bulb more when humidity is low; the temperature difference is compared with a table to find humidity	2
08.4	<i>Any 2 from:</i> - Test strips with reagent bands compared with a colour chart (semi-quantitative) - Electronic colourimeter: reagent added and colour intensity measured at a specific wavelength - Ion-selective electrode measuring electrode potential (other ions can interfere)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	

Question	Marking guidance	Marks
09.1	• Vehicles: high energy density (200 Wh kg^{-1}) minimises battery mass so range is greater; many cycles and high efficiency • Stationary storage: mass is unimportant so the low energy density of lead-acid does not matter; lowest cost per kWh; well-established, recyclable • Lead-acid drawbacks: fewer cycles and lower efficiency; lithium fire risk needs management • Nickel-cadmium largely replaced because of cadmium toxicity and cost	4
09.2	<i>Any 2 from:</i> • Recharging speed • Self-discharge during non-use • Availability of raw materials (lithium, cobalt) • Recyclability / disposal	2
09.3	• Parked electric vehicles plugged into the grid form a very large distributed battery • At peaks a small proportion of their stored energy is fed back to the grid; surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines	2
09.4	• Surplus electricity electrolyses water to hydrogen, which can be converted to methane • Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Stages of scientific method illustrated by ozone: identification (Antarctic UV data), hypothesis (Rowland-Molina 1974), data collection / analysis (Dobson units, satellites, balloons confirming CIO), proposed solutions, enactment (Vienna 1985, Montreal 1987) • Evidence of success: ozone hole area and depth stabilised and beginning to shrink; CFC concentrations in the atmosphere falling; UV levels not rising further; global ozone loss limited to about 4% • Reasons for success: international recognition of serious consequences (skin cancer); agreement of nearly every country; legally binding targets with a fund for developing countries; relatively few producers / products; alternatives (HCFCs, HFCs, pump sprays, hydrocarbons) already available and cheap; no lifestyle change needed • Limits: persistence of CFCs means recovery takes decades; some ODSs still used; HFC replacements are greenhouse gases • Climate change contrast: greenhouse gases come from almost every human activity (energy, transport, agriculture, industry) in every country; fossil fuels underpin economies so control requires major lifestyle and economic change • Agreements (Kyoto, Paris) have weaker / non-binding targets, disputes between developed and developing nations; emissions still rising; natural fluctuations, time delays and tipping points make cause-effect less clear to the public • Judgement: ozone control was a technical substitution problem; climate change is a systemic one, so the ozone success is only partly transferable	9
10.2	• Vienna Convention for the Protection of the Ozone Layer - no legally binding goals	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113, 105-106; AQA spec 3.2.1.3, 3.2.1.2.	

Question	Marking guidance	Marks
11.1	Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Negative feedback: increased low-level cloud (albedo), increased photosynthesis and sequestration; hydrological cycle evaporation-precipitation • Positive feedback: melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • Tipping points: self-sustaining processes; why control must precede them; examples • Implications: urgency and precautionary principle; need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation; geoengineering risks • Difficulty of prediction because feedbacks interact at different scales; models; international agreements and per capita footprints • Judgement on whether current strategies are adequate Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	25

Question	Marking guidance	Marks
11.2	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25